

CELL

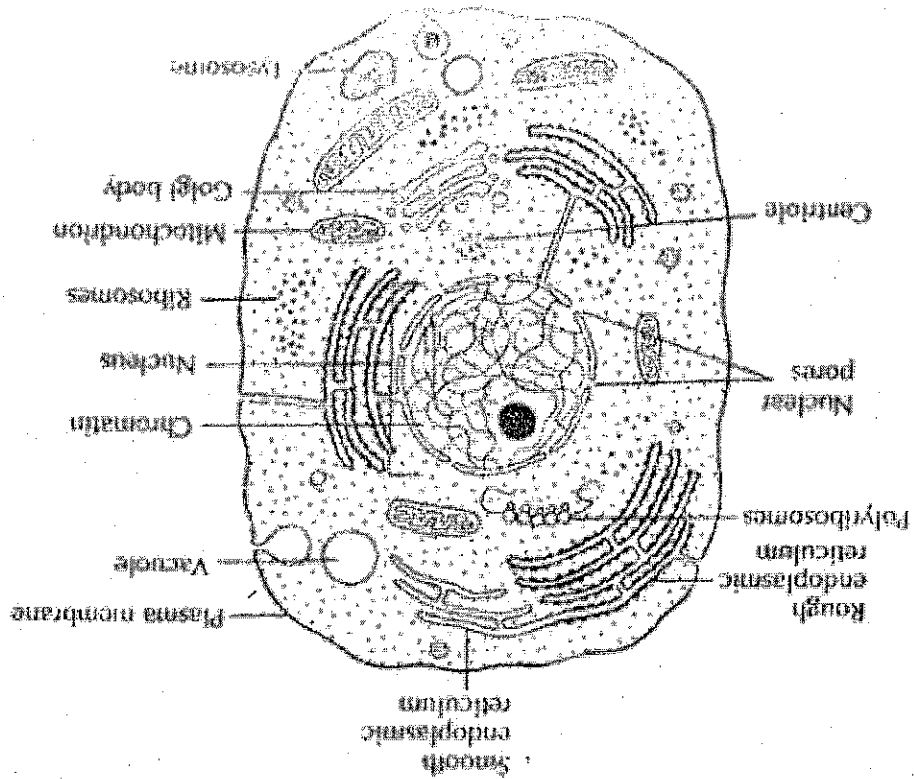
Definition:

The cell is the fundamental morphological and physiological unit of living organisms. The cell contains all the structures and molecular constituents needed for life. Cell of a living organism is divided into two groups: (1) Unicellular (2) Multicellular organisms.

1. Unicellular organisms are those which start their life from a single cell unit and till death continue in the same manner; their body is made up of a single cell. E.g. yeast, diatoms bacteria and protozoans.

2. Multicellular organisms are those which are made up of more than one cell, these organisms also start their life from a single cell but later on it undergoes continued division and develop in to a multicellular body all higher plants and animals are multicellular. The cell ^{of} multicellular organisms get arranged in a definite manner to form different parts of the body. These parts perform different functions and all are governed by the common activity of each cell present in the part of the body.

Electro microscopic studies have revealed that each cell contains in it several living sub-cellular particles called cell-organelles which perform different functions, the chief cell-organelles are mitochondria, endoplasmic reticulum, ribosome, plastids, lysosome, golgi body and nucleus e.g. this it can be said that despite being the unit of living organisms, cell has a much complex structure.



Types of cell

1. Prokaryotic cell

2. Eukaryotic cell

1. Prokaryotic cell (pro = primitive, karyon = nucleus). They are characterized by the absence of nuclear membrane, nucleus, nucleous and most of the well developed cytoplasmic organelles. They are been considered the most primitive type of cells and in them nuclear material is found freely distributed in the cytoplasm, i.e. the nuclear materials like DNA, RNA and proteins are in direct ^{contact} with protoplasm. The chromosome of prokaryotic cell lack histone proteins. Bacteria, viruses, rickettsia, bacteriophages and blue green algae are examples of prokaryotic cell.

2. Eukaryotic cell (greek Eu-good or well, karyon-nucleus). Eukaryotic cells possess all the characters which prokaryotic cells lack. Thus, they are characterized by the present of definitely organized nucleus with nuclear cytoplasm organelles like mitochondria, plastids, ribosomes, endoplasmic reticulum, lysosomes, golgi body etc. nuclear materials like DNA, RNA and proteins in these cell are in contact with karyoplasm. Their chromosomes contain histone protein. E.g algae (except blue green algae) and all higher plant and animal cell.

Osmotic relations of plant cells

Diffusion: The movement of free molecules of gases, liquids and solid from the region of higher concentration to the region of lower concentration due to internal or external forces is called diffusion. Blowing of wind dispersal of good smell of agarbattis in a room, dissolution of sugar in water, ^{intake} intake of CO₂ and ^{liberation} liberation of O₂ in photosynthesis, in take of O₂ and ^{liberation} liberation of O₂ during respiration are all the examples of diffusion. Such a diffusion of molecules in any region contains it spreads throughout the available area. The movement of molecules depends upon the internal kinetic energy. The molecules in the region of higher concentration contain more kinetic energy and that is why they ^{show} slow fast movement. During the movement all the molecules collide themselves and produce a pressure in the medium called diffusion pressure which is directly proportional to the concentration and temperature of diffusing molecules. It means if the temperature and concentration is less, the diffusion pressure will also be ^{less} less.

1. Diffusion pressure and concentration of molecules

2. Diffusion pressure and temperature.

Due to this reason, the diffusion always takes place from the region of higher diffusion pressure (conc.) to the region of lower diffusion pressure.

Mostly gases, liquid and solutes diffuse at different rates in different direction at the same place without affecting each other. This is the reason that O_2 and CO_2 both diffuse in different direction from a single stomata of leaf during respiration. The rate of diffusion is maximum in gases and minimum in solids when dissolved in solvent liquids stand on intermediate position in this regard.

Factors affecting the rate of diffusion

1. Temperature: It is directly proportional to the rate of diffusion. On increasing the temperature, the rate of diffusing particles is also increased because of increase in the velocity of these particles.
2. Concentration of medium: It is inversely proportional to the rate of diffusion. On increasing the concentration of the medium, the rate of diffusion is reduced and on decreasing the concentration, it is increase.
3. The size and mass of the diffusion particles: If the size and mass of the diffusion particle is smaller, the rate their diffusion will always be faster.
4. Solubility of solutes: The rate of diffusion increase with the rate of dissolution of solute in solution. Thus, the more the solubility of any solute in solution, the faster will be the rate of diffusion.
5. Diffusion pressure gradient (DPG) : The rate of diffusion molecules of gases and liquid also depends upon the diffusion pressure gradient (DPG). When the difference in the diffusion pressure is more, faster is the rate of diffusion.
6. Density of the diffusion particles: The rate of diffusion of gases is related with the density of diffusion particles. According to Graham's law of diffusion, "the rate of diffusion of any gas particle is inversely proportional to the square root of its density.

$$r = \frac{1}{\sqrt{d}}$$

Where: r = rate of diffusion of a gas
d = density of the gas

The gases having higher densities shows slower rate of diffusion where as those possessing lower density slower rate of diffusion. For example, the density of hydrogen (H) is one and of oxygen (O) 16. Therefore, according to Graham's law of diffusion:

$$\frac{r}{H} = \frac{\sqrt{d_0}}{\sqrt{d_1}} = \frac{H}{\sqrt{d_1}} = \frac{1}{4}$$

Thus, the hydrogen will diffuse four times faster than oxygen.

Importance of diffusion in plants

- i. The exchange of CO₂ gases in the atmosphere through stomata of leaves takes place by the process of diffusion. O₂ gas participates in respiration whereas CO₂ in photosynthesis.
- ii. During stomata transpiration the water vapors from intercellular spaces diffuse in the atmosphere through stomata by the process of diffusion.
- iii. The diffusion of ions of mineral salt during passive absorption takes by this process.
- iv. The absorption of water through roots is also performed by diffusion.

Permeability: The entrance of exit of any substance in the cell depends upon the permeability of plasma membrane. The permeability of plasma membrane is usually changes from times to time due to changes in either K⁺ ions concentration of nature of other particles surrounding it. Depending upon the permeability properties the membranes can be classified into following categories.

- (a) Impermeable: these are those membranes through which of only the following exchanges of only gates takes place.
- (b) Semi-permeable: these are permeable through which exchange of only water or solvent molecules takes place. Solute particles neither can enter nor leave through of goat, and egg membranes are the examples of semi-permeable membrane.
- (c) Selective permeable: These are the membranes through which the exchange of only selected ions or small molecules takes place. Plasma membrane is mainly selective permeable and not only semi-permeable.
- (d) Dialyzing membranes; These membranes possess other layers in the form of water covering.

Plasma membrane is usually permeable for gases like CO₂, O₂, N₂, solvents like alcohol, either chloroform and organic substance like mono- and disaccharides, fatty acids, amino acids and glycerol whereas it is normally impermeable for polysaccharides and proteins. The permeability of plasma membrane for gases and solvent is more than organic substances.

Factors affecting permeability

A number of external and internal factors affect the permeability of plasma membrane.

External factors

1. Physical agents: the physical agents like high temperature, heat, low pressure of O_2 and CO_2 , radiation etc. increase the permeability of plasma membrane.
2. Chemical agents: a number of chemicals like ether, benzene, chloroform, acetone, toxic substances etc. when added in the external medium or solution increase the permeability
3. Ageing of cells: The permeability ^{varies} at different ages of the cell. In young cells, it is low but at the times of senescence it increases the older cells approaching death also shows increase in permeability.

Internal factors

V Membrane Constitution: the permeability of plasma membrane depends upon its constitution, i.e. percentage of lipids, proteins, carbohydrates and phosphate etc. thickness, porosity and degree of hydration.

Osmosis:

When two solutions of different concentrations are separated by semi-permeable membrane, the diffusion of water or solvent molecules takes place from the solution of lower concentration towards the solution of higher concentration. This process is called osmosis.

Actually the diffusion of solvent molecule takes place across the membrane on both the sides but it is faster from the lower concentration side than the higher concentration side. This is according to principles of diffusion because the solution of lower concentration possesses higher concentration of solvent molecule whereas the solution of higher concentration possesses comparatively lower osmosis is a type of diffusion through semi-permeable membrane.

From biological point of view, the solutions may be of following three types:

1. Hypertonic solutions: those solutions whose concentration and osmotic pressure (OP) are more than the concentration and OP of cell-sap are called hypertonic solutions.
2. Hypotonic solutions: Those solutions whose concentrations and OP are less than the concentration and OP of cell-sap, are called hypotonic solutions.
3. Isotonic solutions: Those solutions, whose concentration and OP are equal to the concentration and OP of cell-sap, are called Isotonic Solutions.

Types of Osmosis: osmosis in plants is of two types:

following two types:

According to Renner (1912 and 1915), the mechanism of water absorption is of

Mechanism of water absorption

Abortion of water

S/N	Diffusion	Osmosis
1	Diffusion occurs in solid, liquid and gases	Osmosis occurs only in liquid
2	Does not require semi permeable membrane	Requires a semi permeable membrane
3	In this process, the movement of molecules takes place from higher concentration	The movement of water or solvent molecules takes place from the solution of lower concentration to the solution of higher concentration

Differences between diffusion and osmosis

1. The water is most important factor for life because all the biological activities are directly or indirectly controlled by water. Osmosis helps the plant in the absorption of water from the soil through roots.
2. The turgidity in plant cell is maintained by the absorption of water through osmosis.
3. The diffusion of water from one cell to another and its distribution in different plant parts is also performed by osmosis.
4. The opening of stomata depends upon the turgidity of guard cell which is maintained by osmosis.
5. Osmosis pressure and turgidity or pressure helps in the growth of young cells.
6. High osmosis concentration in plants increases the resistance against freezing temperature and dehydration.
7. Osmosis helps in the dehiscence of fruits and sporangia.

Significance of osmosis in plants

- i. Endosmosis: when water or solvent molecules enter into the cell through plasma membrane from the outer medium, it is called endosmosis.
- ii. Exosmosis when a plant cell is placed in concentrated solution, the water molecules move from cell into the outer concentration medium through plasma membrane. It is called exosmosis. Thus, in exosmosis the water or solvent molecules leaves the cell into the outer medium through plasma membrane.

1. Active absorption: when roots are involved actively in water absorption and water absorbing forces in plants are developed primarily in roots, such types of absorption are called active absorption. It is found in those plants where transpiration is less and water is present in sufficient amounts. Active absorption requires ATP released during respiration. It is also of two types:

(a) Osmotic absorption: in this type of absorption the roots act like osmometer and water absorption according to osmotic gradient.

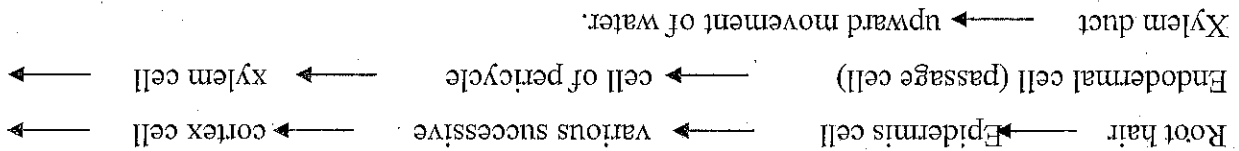
(b) Non-osmotic absorption: in this type of water absorption more ATP is required and water absorption takes place against osmotic gradient.

2. Passive absorption: when roots are in active in water absorption and the water absorption forces develop primarily in leaves and stem and then reach to root through xylem, this type of water absorption is called passive absorption. It takes place mainly due to transpiration. Passive absorption is found in those plants where transpiration is very fast and it does not require ATP.

3. ^{osmotic} Osmosis active absorption: soil is made up of irregularly arranged soil particles. Well developed air spaces are found among those particles. The spaces contain capillary water where air and mineral salts remain dissolved. The roots having remain spread in this water. It was already been described that each root hair contains a vacuole which remains filled with mineral salts, sugar and organic acids. The cell-wall of root hair is permeable and it does not create any hindrance in the entry and exit of the liquids in the cell. Plasma membrane is semi permeable and it allows only the different. The cell-wall of root hair being hydrophilic in nature first absorbs soil water through ion bibityon. The cytoplasm of root hair is usually concentrated than the capillary water of soil.

Pathway of water in root

The pathway of water in root cell is as follows:



ROOT PRESSURE: when the water enters from turgid pericycle cell into xylem vessels, a pressure is created in the xylem of roots due to which the water rises to a certain height in the xylem. This pressure is called root pressure. Actually the root pressure is a type of hydrostatic pressure which is produced in the cell-sap of xylem vessels.

Theories regarding development of root pressure

Following theories have been proposed to explain how the root pressure develops in plants;

1. Secretion theory: according to this theory. The root pressure develops by the secretion of water into the xylem. The secretion of water results due to difference in the permeabilities of water in two sides of the root cells. It is higher towards inner side and lower towards outer side. Some energy may be required for this work which is provided by respiration through root cell.

2. Electro-osmotic theory: according to this theory the root pressure develops due to electro-osmosis movement of water into the xylem. It has been proved experimentally that when an electric current is passed into the membrane, water moves across the membrane. However, this theory was seen discarded.

3. Osmotic theory: according to this theory the root pressure develops due to osmosis. The root behaves like an osmometer due to accumulation of xylem gap. Thus, water moves from soil solution to xylem through various tissues of root developing root pressure.

Non-osmotic active absorption

According to Thimann (1951) and Kramer (1959) the water absorption is an active process which takes place against the osmotic gradient. In other words, sometimes the water absorption also takes place when the osmotic pressure of soil water, is greater than the osmotic pressure of cytoplasm of root hair. Thus is against osmosis. Such type of water absorption is called non-osmotic active absorption and it requires ATP produced during respiration of root cells. How this energy is utilized, it is not well worked out. The energy may be utilized directly, or indirectly. The theory of non-osmotic active absorption is supported by the following facts.

- i. Correlation between water absorption and rate of respiration usually the presence of respiration reducing factors the rate of water absorption is also reduced. This, both the process inter-related

- ii. Reduction of water absorption in presence of respiratory inhibitors: it has been observed when plant is placed in the solution of respiratory inhibitors like KCN, the rate of water secretion is also reduced or stopped. This indicates that respiration and water absorption processes are inter-related. KCN directly, destroys the coenzymes of respiratory chain like NAD, FAD and cyto-chromes and thus effects respiration and other metabolic activities

- permeability.
- iii. It increases the concentration of protoplasm which result in the reduction of
 - ii. It reduces the size and growth of roots
 - i. CO_2 reduces metabolism activities and respiration rates
- accumulation of CO_2 which effect the plants in following manner.
- In other words, the rate of water absorption is increased is due to lack of O_2 and
2. Soil aeration: The rate of water absorption increases when the aeration in soil is high.
- absorption. Similarly, on shortage, the water absorption rate is also reduced.

capacity, it creates a bad effect on soil aeration and also reduces the rate of water permanent wilting percentage. If the amount of water is increased than the fields absorption in plants remains constant the same in between field capacity and only capillary water is available for the absorption of plants. The rate of water (gravitational, Capillary, Hygroscopic and chemically bound water) but out of these 1. Available soil water: It has been discovered that soil water is found in many forms

Factors affecting the rate of water absorption

S/N	Active absorption	Passive absorption
1	It requires ATP ATP	It does not require energy
2	It creates root pressure	Root pressure is not created due to P.A (passive absorption)
3	It requires oxygen	It does not requires oxygen
4	The movement of higher concentration of the solution of lower concentration i.e. against the concentration gradient	The movement of water takes place from the solution of lower concentration to the solution of higher concentration i.e. according to osmotic gradient

Differences between active and passive absorption

- iii. Wilting of plants in oxygen deficient soil: in O_2 deficient soil, the plants do not get sufficient O_2 for respiration and ultimately *die*.
- iv. Effects of auxins: in presence of auxins, the rate of metabolisms along with water absorption are increased.
- v. The occurrence of water absorption and respiration only in living cells. Both these processes are found only in living plants and not in dead plant.

- Due to above effects of CO_2 , the rate of water absorption is decreased. This situation is created in water-logged soil where the aeration is quite poor. Such soil is usually physiologically dry and is not fit for water absorption.
3. Concentration of the soil solution: the osmotic pressure of any solution is directly proportional to its concentration. In other words, the OP is more concentrated solutions. If the soil water containing more salts, its concentration and OP will also be more. The rate of water absorption will be reduced. If the OP of soil solution will be more than OP of cell sap of roots. Due to this reason, the plants growing in alkaline and marsh soil shows a very little or no water absorption.
 4. Soil temperature: generally, the rate of water absorption is maximum between $20-30^\circ\text{C}$. it is reduced if the temperature is less than 20°C or more than 30°C . at very low temperature or at 0°C , the water absorption is almost stopped and its rate of water absorption is affected directly or indirectly in the following manners:
 - i. It increases the concentration of protoplasm
 - ii. It reduces the permeability of plasma membrane
 - iii. The growth of root is reduced which result in reduction of their length
 - iv. The rate of diffusion of soil-water into roots is also reduced
 - v. Metabolic activities of root cells are also reduced
 5. Root system: root system also affects the rate of water absorption. Those plants which possess hairy and well developed root system flows higher rates of water absorption in comparison to those possess very small roots and fewer members of root hairs.
- ### Importance of water to the plants
- Although water plays many roles in plants but some of the important ones are:
1. The amount of water present in the soil changes the morphology and anatomy of the plants. Accordingly the plants are mesophytes, hydrophytes and xerophytes.
 2. Water is good solvent for many substances such as organic salts, sugars and organic anions i.e. it dissolves other substances.
 3. Water helps in the formation of protoplasm.
 4. The absorption and translocation of mineral salt and dissolved substances take place through water.
 5. Water is an essential raw material in photosynthesis. During photosynthesis, water is oxidized and O_2 is produced
 6. Water affects transpiration seed germination, respiration, dispersal of fruits and seeds, activation of enzymes, hydrolysis of ATP growth and all other metabolic processes.

7. Water maintain the temperature in plant tissues
8. Water maintains the turgidity of plant body

Soil water

Soil water is the most important constituent of soil because it provides the medium of the absorption of minerals and organic matter by the roots and also activities various enzymes and metabolic processes. It affects morphological and physiological processes directly or indirectly. Soil-water later on reaches to the various other parts of the plant through roots. The vegetation of an area depends upon the quality of soil water. The water present in the soil may be of following types:

1. Gravitational water: The water which reaches deeply into the soil after rains due to gravitation is called gravitational water. The plants cannot absorb this water through their roots.

2. Surface tension: Is the forces which exists as a combination of cohesive force between water molecules, and adhesive force between water molecules and sand particles. It is this force which makes water drops spread on the surface of soil particles. It also enables water to rise up through narrow pores.

3. Capillary water: Is water which rises above the water table in the soil and is held in the fine and medium pores of soil particles by surface tension. Capillary water moves from more saturated to less saturated regions. The finer the pores, the greater is the force binding the water (i.e surface tension), and the higher is the capillary tension. This results in a higher rise of water. Capillary water is the only available water which is absorbed by plant roots.

4. Hygroscopic water: Is that water which is absorbed on the surfaces of soil particles from atmospheric water vapour. The colloidal components of the soil, such as clay and humus, have a strong attraction for water molecules, binding them to their surfaces by surface energy forces. The amount of water absorbed from the atmosphere is always in equilibrium with atmospheric moisture. It is non-available water and is not absorbed by plant roots.

5. Crystalline water or chemically combined water: The water which remains chemically bound to the soil particles is called crystalline water. It is also non-available water, e.g. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, etc.

6. Running water: The non-available water which after rains flow down through the slopes is called running water.

is always in equilibrium with atmospheric moisture. It is non-avaliable water and is not absorbed by plant roots.

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Forms of underground water

Underground water is found in following two forms:

1. Water table: It is found quite deep in the soil near rocky layer. It contains free water which comes down into the soil through germination. The water from water-table ascends/towards soil-surface through capillary forces

2. Capillary fringe: It is present first above the water table and represents that part of soil where the water ascends from water-table by capillary forces. Capillary fringe is absent in gravel soil. The soil portion which is present above the capillary fringe is called aeration zone. It contains more field-capacity. The field capacity of any soil is the percentage of water present in that soil which remains after the gravitational water is drained away. Thus, field capacity of any soil is the sum of capillary water, hygroscopic water and crystalline water.

Field capacity = capillary water + hygroscopic water + crystalline water.

Water content or hygroscopic coefficient

The percentage of water present in terms of dry weight of the soil is called hygroscopic coefficient.

It can be calculated by collecting soil samples from the fields, weighing them in balance and drying in an oven at 105°C to obtain constant weight. The loss in weight in the soil sample will be water content or hygroscopic coefficient of that soil

Moisture equivalent

The percentage of water a soil can be held in opposition to pull 1,000 times that of gravity is called moisture equivalent.

Wiltng

In dry atmosphere, sometimes the plants shows scarcity of water in them and their leaves become wilted. It is called wiltng. There are two reasons for it:

The part of the plant which develops from the radical of germinated seed is called root. At the time of seed germination, first radical emerges which forms primary root. The secondary and tertiary roots are formed from primary and secondary roots respectively. Primary, secondary

Root system in plants

- i. Dry or wet surface of leaf
- ii. Presence of hairs on the leaf-surface
- iii. The structure of epidermis and cuticle and their permeability
- iv. Lack of water in the cells of cortex

Following factors affect water absorption by leaves:

The absorption of water in different types of plant is performed by some of their special organs. In lower plants like aquatic fungi and algae, the water is absorbed by their all cell of plant body as they are found submerged in water. In other aquatic plants, the absorption of water takes place by the cells of external surface of different organs. Such plants do not have any special organs for water absorption. In liver worts and mosses the water absorption takes place through unicellular and multi-cellular rhizoids which are similar to those of roots hairs present in higher plants and also perform the function of roots. Liver worts possess unicellular rhizoids whereas mosses possess multicellular rhizoids. In pteridophytes, gymnosperms and angiosperms the water is mainly absorbed by the roots present in the soil. These roots remain spread in the soil. The roots possess many unicellular root hairs which absorb only capillary water of the soil. The epiphytes like orchids, possess aerial roots which contain valance layer. The cells of this layer are hygroscopic in nature and absorb water present in the atmosphere. In some plants like sorghum, water is absorbed by leaves. The

Water absorbing parts of the plants

- i. High rate of transpiration by plant in comparison to less absorption of water by roots. The wilting due to this reason is called temporary wilting because the leaves require their natural condition during night due to low transpiration rate.
- ii. Less quality of water in soil: When the leaves become wilted due to less quality of water in soil, it is called permanent wilting. In such stage, the leaves do not regain natural condition at night. The permanent wilting can be overcome by adding water in the soil.
- Permanent wilting point: The soil moisture content present at the time of permanent wilting is called permanent wilting point
- Wilting coefficient: The percentage of water present in soil after permanent wilting is attained is called wilting coefficient

1. Epidermis: It is single-layered, hairy and without stomata and cuticle.

Following structures are seen in internal structure of root:

Internal structure of root

- additional roots which provide anchorage to the plant.
- e. Mature region: The region is present above the region of root hair. It produces many hairs. Utricularia, Hydrilla and Spirogyra etc. do not possess root hairs
 - d. Region of root hair: It is found just above the region of elongation and possess many unicellular root hairs. Here, newly-formed meristematic cells are converted into permanent tissues. Root hairs are important organs of plant because they help in the absorption of water and mineral salts. One millimeter area possess about 2000 root hairs.
 - c. Region of elongation: It is found just above the meristematic region in about 2-5mm in length. In this region, the cells elongate resulting in increase in length.
 - b. Meristematic region: It is also known as region of cell division and region of growth. It is found just above the root cap region in about 1mm in length. It remains covered by many root hairs. The cells of this region are active, thin cell-walled and with large protoplasm. They divide and redi-vide to form new cells.
 - a. Root cap region: It is found at the tip of root. It protects the root from injury at the time of growth. During growth when old cells become dead, new cells are formed.
- A root possesses the following distinct five regions:

Regions of the root

7. Root possesses many unicellular root hairs which help in the absorption of water and mineral facts.
6. Root possesses a root cap at their tips which protects the roots from injury at the time of growth.
5. Roots lack buds, in some plants the roots possess negative buds which develop into new plants.
4. Roots lack nodes, internodes, leaves, fruit and flowers.
3. They are usually brown, white or ash colored.
2. Roots are negatively phototropic.
1. Root is the descending part of the axis which develops from the radical of embryo.

Characteristics of root

and tertiary roots collectively form the root system. In some plants the special roots are formed which do not develop from radical.

The roots of plants absorb water and mineral salts continuously. The total amount of water absorbed is not retained in the plants but only a very small amount necessary for various activities and composition of various organs of plants is retained. The excess amount is transpired through aerial parts (stem, leaves, flowers, and buds) of the plant. Thus, only 5 percent of the absorbed H_2O is retained in the plants and remaining 95 percent is lost through aerial parts. The leaves are most important for transpiration. The loss of excess water in the form of vapor from the various parts of the plants is called transpiration. The transpiration is like evaporation but both differ from each other in following points.

TRANSPIRATION

Root hairs develop from the epidermal cells in the form of outgrowths. Each root hair is unicellular. Its cell-wall is made up of cellulose and pectic substances which are hydrophilic in nature and are capable of absorbing water. Cell wall is permeable. Plasma membrane is found first below the cell-wall. In the centre of the cell, a large vacuole is found which contains minerals salts, sugars and organic acids in the dissolved form. It controls the absorption of water. The space between plasma membrane and vacuole contains cytoplasm, nucleus and many small vacuoles.

Structure of root hair

8. **Pith:** It is either absent or very small.
- parenchyma.**
7. **Phloem:** It is usually made up of sieve tubes, companion cells and phloem towards the centre.
6. **Xylem:** It is exarch, protoxylem being present towards the periphery and metaxylem
5. **Vascular bundles:** They are radial and remain arranged in a ring.
4. **Pericyle:** It is single-layered, parenchymatous and without intercellular spaces.
- Endodermis** possess passage cells and calparian strips.
3. **Endodermis:** It is well developed, single layered and made up of barrel shaped cell. intercellular air spaces.
2. **Cortex:** It is multicellular, parenchymatous and with many well developed

Differences between transpiration and evaporation

S/N	Transpiration	Evaporation
1	Transpiration is biological phenomenon	Is a simple and physical phenomenon
2	It is controlled by guard cells	It is not controlled by guard cells
3	It takes place through surface of leaves	It takes place through the surface of various water bodies and does not require the living organs of plant like leaves
4	It takes place due to osmotic pressure and suction pressure	Osmotic pressure and suction are not required
5	It takes place in living cells	Plant cells are not required
6	Maintain plant temperature	Has no relation with plant temperature

Kinds of transpiration

The process of transpiration takes place through all aerial parts of the plant but through leaves it is maximum. Transpiration may be of following types:

➤ Cuticular Transpiration

The loss of water in the form of vapor from the cuticle of various aerial parts is called cuticular transpiration. The water vapor reach to the cuticle through the internal tissues of aerial parts by diffusion and from cuticle it diffuses in the atmosphere of through cuticle. The plants growing in shady places transpire up to 15 percent of the total transpired water. the amount of water loss in xerophytic is quite little or negligible. In flowers, stems and fruits mainly cuticular transpiration is found.

➤ Lenticular Transpiration

The loss of water in the form of vapor through lenticles is called lenticular transpiration. Lenticels are the small pore present below the bark of old trees.

➤ Stomatal Transpiration

The loss of water in the form of vapor through the stomata of leaves is called stomatal transpiration. The maximum amount of (80-90 percent) absorbed water is transpired through stomatal transpiration. Stomatal transpiration is commonly found in the leaves and stems of young plants. The crop plants like wheat, grain, maize, mustard and potato etc. also transpire through stomata. The water loss 250 tonnes/day acres has been recorded from wheat crop.

STRUCTURE OF STOMATA

The epidermis of leaves and stems possess many pores called stomata (singular: stomata). The length and beath of open stomata is about 10-40µ and 3-10µ respective.

Each pore remains surrounded by special semi-lunar or kidney shaped living epidermal cells called guard cells. The pore and guard cells jointly called stoma. The stoma opens to the interior into a cavity called substomatal cavity which remains connected with the intercellular spaces.

The guard cell possesses a nucleus, cytoplasm and several chloroplast. Their inter walls are thicker than the water walls. Each guard cell is surrounded by two or more living cells called subsidiary or accessory cells.

Distribution of stomata on leaf

The stomata are found distributed on the upper and lower surface of the leaves. In dorsiventral leaves, their number is more on the lower side in Isobilateral, their number is equal on the lower and the upper both the surface. In floating leaves e.g. water-lily, the stomata are found only on the upper surface. In submerged plants like Hydrilla and potamogeton non-functional stomata are found. In xerophytes like copparis decida, casuarina and phichia lanceolata sunken stomata are found in Nerium and Pinus, the stomata are found sunken on the cavities.

In dicotyledonous leaves usually the stomata are found scattered where as in monocotyledonous leaves they are arranged in parallel rows. The number of stomata in a leaf varies from 1000 to 6000 per square centimeter. All the stomata of a leaf cover about 1-2 percent of total area of leaf.

Types of stomata

The stomata may be classified into following types according to their distribution of leaves:

- i. Apple type: when the stomata are found only in the lower surface of the leaf, the condition is called hypostomatous. The condition is found in Apple, peach Mulberry and walnut.
- ii. Potato type: when the stomata is only found more on the lower surface than the upper surface e.g. in potato, tomato and pea.
- iii. Oat-type: when the stomata are found equally on both the surface e.g. oat, maize and grasses
- iv. Water-lily type: when stomata are found only on the upper surface e.g. water-lily
- v. Potamogeton type: when the stomata are absent or non-functional e.g. potamogeton.

PLANT ANTI TRANSPIRATION

Those substances which reduce the rate of transpiration in plants are called plant antitranspirants. Some of the main antitranspirants are colorless mercuric acetate, abscisic

acid, CO₂, oxy-ethylene, and decently succinic acid. They retard the transpiration without affecting CO₂ fixation in photosynthesis or plant growth.

Factors affecting transpiration

Following factors affect the transpiration (a) external or environmental factors, (b) internal or structural factors

External on environmental factors

1. Humidity in air: wet and dry air affects the transpiration. The air absorb moisture. If the air of the atmosphere is dry and the air present inside the intercellular spaces of leaves is wet, the diffusion of water-vapor will take place from less humidity in the air decreases the rate of transpiration.

2. The rate of transpiration is directly proportional to the temperature of atmosphere. *It* also affect the humidity of air. The rate of transpiration increase with the increase in temperature and decreases with the decrease in temperature of the atmosphere. The high temperature of atmosphere lowers and low temperature increases the humidity. 3. Light: transpiration increases in high light intensity and decrease in low intensity. In high intensity of light, the water present in mesophyll cell diffuses rapidly resulting in increase in humidity of internal air. This increases the rate of transpiration. The stomata remain open during day hours and closed during night hours.

4. Wind velocity: high wind velocity increases the rate of transpiration and low velocity decreases the rate of transpiration. In high velocity, the humid air are surrounding the leaves moves all the sideways from the plate, resulting in increase in humidity around the leaves and increase the rate of transpiration.
5. Water content in soil: if excess water is present in the soil, more absorption of water will take place resulting in high rate of transpiration. In water deficient soil, the rate of transpiration decreases.

Internal factors

1. Presence of cuticle on epidermis: some leaves possess thick cuticle on the epidermis. It reduces the rate of transpiration.
2. Hairy leaves: some leaves possess epidermal hairs which retain moisture or humid air resulting in retardation in transpiration
3. Presence of sunken stomata: the nerophytic leaves possess sunken stomata which also reduces the transpiration rate

4. Number of stomata on leaves: the rate of transpiration is related with the number of stomata. More number of stomata present on the leaves, more will be the transpiration rate and less number of stomata show lower transpiration rate.
- plants*
- Special features in plants to reduce transpiration**

The following three types of features are found in plants to reduce transpiration:

a. Morphological features

b. Anatomical features

c. Physiological features

Morphological features

1. Presence of dry, hard and cylindrical stem e.g., salvadora, and leptaedaenia. Some stems possess midges and furrows e.g. casuaniva
2. Presence of bark and cork on stems e.g. Acacia
3. Reduction in number of branches and size of stems. Some stems become flat and fleshy e.g. Opuntia.
4. Decrease in length and number of branches in roots. They reach deeply in the soil. E.g. Alfalfa

5. Presence of fleshy roots for storage of water e.g., Asparagus
6. Presence of small scaly and reduced leaves Ruscus and Tamarix.
7. Division of leaves into many small leaflets or lamina e.g. Acacia
8. Presence of long marrow and needle like leaves e.g. pinus
9. Presence of small scaly and reduced leaves e.g. Ruscus and Tamaxis
10. Presence of thick leaves covered by thick waxy layer or cuticle e.g. calotropis, ficus and salvadora.
11. The leaves of some plants become thick, fleshy and mucilaginous or possess resin or gum e.g. Aloe and Agave

12. Modification of stipules into thorn e.g. ziziphus and Acacia
13. Presence of excess hair or hairy covering on the leaves, flowers and stems e.g. Aerva, calotropis

14. Rolling and folding of leaves e.g. Marrum grass

15. Presence of smooth and flining leaves e.g. Neninum

16. Production of gum, resin and mucilage etc. in the cells of plants e.g. pinus, Opuntia and cypas

Anatomical features

1. Presence of thick cuticle e.g. Agave, Dianthus and fiscus

GUTTATION: When conditions are such that the absorption of water greatly exceeds transpiration, the excess of water escapes through structures called **hydathodes** present at the tips of veins of leaves water may, sometimes, escape from the stems the scars of leaves and lenticels. This type of exudation is called guttation. In other words, guttation is the water exudates through hydathodes in the form of liquid droplets. The liquid coming out of hydathodes is not pure water but a solution containing a number of dissolved substances like P, K, Na, Fe, Mg, glucose, fructose, sucrose, thiamine and riboflavin.

Other methods of water loss

1. The water and minerals absorbed by the roots from the soil reach continuously in different parts of the plants through transpiration
2. Transpiration maintains the concentration of mineral salts
3. Transpiration maintain the temperature of plants
4. Transpiration increases the SP of mesophyll cells of the leaves which results in increase in the absorption power of roots.

Importance of transpiration to plants

1. Presence of high OP in the cell-sap of leaves
2. Presence of hydrophilic substances in the cells of stems and leaves which retain water and becomes fleshy
3. Presence of excessive growth in the roots due to which they become quite long and reach at a greater depth to absorb the water e.g. Allagi cameloram
4. Closing stomata during adverse conditions of environment

Physical features

10. Formation of cork layer in the stem
9. Formation of cork and bark e.g. Acacia, Zizphus and prosopis
8. Presence of compactly arranged palisade cells e.g. cya ad nenim
7. Presence of excess amount of scalerenchyma e.g. salvadora
6. Presence of mucilage, gum resin, latex and tannin in the cells of hypodermis and cortex e.g. opuntia, pinus, flais and cycas
5. Presence of sunken and less number of stomata e.g. most of the xerophytes and sometimes in the depressions of epidermis e.g. Nenum
4. Presence of hairs and scales on the epidermis
3. Presence of multilayered epidermis e.g. Nerium, fiscus and pepromia
2. Presence of another waxy layer on the cuticle e.g. calyx glaucophylla

Growth is an important character of the all living beings. Each living organism increases in size, shape, volume and weight. This process is called growth. It requires nutrition. Physiological processes play important roles during the growth of plants and animals. Growth is always causes due to internal forces, during which the cells divide, increase in length and

GROWTH AND METABOLISM OF GROWTH HORMONES

BLEEDING: The exudation of water and cell-sap through the cuts or wounds of plants is called bleeding. It also happens due to positive root pressure. In certain plants the pressure is developed either in the phloem elements or in the cells surrounding the ^{cut} or wounds. The exudation of toddy juice from the wounded stem of Borassus. Toddax and latex from Haevea (rubber) stem are the examples of bleeding. It has been utilized in the production of various economically important products like, rubber, sugar, and Alcoholic drink.

S/N	Transpiration	Guttation
1	H ₂ O takes place in form of vapor	H ₂ O loss in form of liquid
2	H ₂ O is pure water	Liquid contains a number of dissolved substances e.g. mineral salt, sugar, coenzymes
3	Takes place through stomata, lenticles and cuticles	Takes place through hydathodes which found at the margins of leaves
4	Stomata usually remain open during day and closed during night	The hydathodes remain open whole day and night
5	It takes place during day hours	It takes place either in the morning or during night
6	It is controlled by the guard cells	It is not controlled by guard cells
7	Root pressure is not involved	Root pressure is involved
8	Maintains the temperature of plant	It has no relation with temperature
9	It takes place in all the higher terrestrial plants	It takes place mainly in herbaceous plants e.g. tomato, grasses.

Differences between transpiration and guttation

It has been pointed out earlier that the stomata becomes closed on sunset and remain open whole day and weight (24 hours). The hydathodes remain connected with the veins. Each hydathodes open to the exterior by means of a pore called stomata. The stomata is surrounded by two guard cells opens internally in to an air chamber. Just below the air chamber there is a group of parenchymatous cells with intercellular spaces called epithem. The xylem vessels and tracheids open just behind drops takes place through stomata of hydathodes. The liquid moves from xylem elements into phenomenon of guttation takes place due to development of positive root pressure.

for each plant or the light received during day time by a plant is called photoperiod.

- i. Light: It chiefly affects photosynthesis and transpiration. The favorable length of day

The growth of plants is affected by following external and internal factors:

FACTORS AFFECTING THE GROWTH OF PLANTS

- i. Increase in length or girth as in case of stem and roots
- ii. Increase in weight
- iii. Increase in volume or area as in case of fruits and leaves respectively
- iv. Increase in number of cells per unit area

The growth of a plant can be measured on the following basis:

MEASUREMENT OF GROWTH

1. Lag period of growth: During this period, the growth rate is quite slow because it is the initial stage of growth. In other words, the growth starts from this period.
 2. Log period of growth: During this period, the growth rate is maximum and reaches at the top because at the stage the cell division and physiological process are quite fast.
 3. Decline period growth: This period comes after long period. During this stage the growth rate slows down because the metabolic process becomes slow.
 4. Senescence period or steady period: During this period, the growth is almost complete and becomes static thus, the growth rate becomes zero.
- The growth rate of a cell, a plant organ or a whole plant does not remain in the same but always varies. This variation in growth is due to irregular changes. In initial stages during the phase of cell formation, the growth rate is quite slow while it increases in rapidly during the phase of cell elongation and becomes maximum, later on it slows down during the phase of cell maturation. Of the last, a stage comes when the growth rate becomes zero i.e. the growth stops. Thus, the total period from initial to final stage of growth is called grand period of growth. The total period of growth can be divided into following stages:

Course of growth

finally differentiate into different types of tissues. In animals, the growth stops after definite stages while in plants it continuous during the entire life. In lower plants, it takes place usually in the entire body while in higher plants, it takes place mostly at the apex of different organs like root, stem and leaves because the meristems remains situated at this part. They are called primary meristems. In the older regions of these organs secondary meristems like cambium and cork-cambium arise which cause secondary growth. The non-living materials like stones and crystals of copper sulphate in its solution also increase in size but this growth always takes place by the deposition of particles of these substances through external forces.

The response of plants to the relative length of the day and night is termed as photoperiodism. According to the light duration, the plant may be short day, long day and day neutral.

2. Temperature: The plant growing in different regions require different temperature for growth. The temperature has a pronounced effect on growth. It occurs between 4°C to 45°C . The best growth takes place between 38°C to 44°C .

3. Oxygen: It is necessary for respiration during which the food materials are oxidized to release energy. The growth is directly proportional to the amount of oxygen. In excess of oxygen, the growth is more while it is reduced when amount of O_2 is reduced.

4. Carbon dioxide: CO_2 chiefly affects the process of photosynthesis. Its rate increases in excess of CO_2 and results in the manufacture of food materials like carbohydrates etc. these food materials are most important for growth. The rate of photosynthesis is slowed down when the amount of CO_2 is not sufficient.

5. Water: The water is most important factor for growth in its presence various physiological activities like water absorption, translocation of food materials, photosynthesis, activation of enzymes and protoplasm etc. take place.

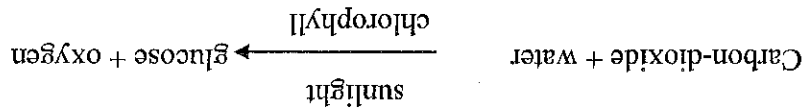
6. Food materials: The growth rate increases in the presence of excess food materials while decreases during shortage. They also increase the concentration of cytoplasm and rate of cell-division.

INTERNAL FACTORS

Internal factors necessary for growth mainly include growth-hormones like auxins, gibberellins and cytokinin etc.

PHOTOSYNTHESIS

Photosynthesis is the process by which plants, some bacteria and some protists use the energy from sunlight to produce glucose can be covered into pyruvate which releases *calories* adenosine triphosphate (ATP) by cellular respiration. Oxygen is also formed. Photosynthesis may be summarized by the word equation:



The conversion of usable sunlight energy into chemical energy is associated with the action of the green pigment chlorophyll is a complex molecule. Several modifications of chlorophyll occur among plants and other photosynthetic organisms. All photosynthetic organisms have chlorophyll a. Accessory pigments absorb energy that chlorophyll a does absorb accessory

pigments include chlorophyll b (also c, d and e in algae and protoplasts), xanthophylls, its energy from the violet-blue and reddish orange-red wavelengths, and little from the intermediate (green-yellowish orange) wavelengths.

All chlorophyll have:

- A lipid-soluble hydrocarbon tail ($C_{20}H_{39}$)
 - A flat hydrophilic head with a magnesium side-groups on the head
- The head and the tail are linked by all ester bond

LEAVES AND LEAF STRUCTURE

Plants are the very photosynthetic organisms to have leaves (and not all plants have leaves). A leaf may be viewed as a solar collector crammed full of photosynthetic cells.

The raw materials of photosynthesis, water and carbon dioxide, enter the cells of the leaf, and the products of photosynthesis, sugar and oxygen, leave the leaf.

Water enters the root and is transported up to the leaves through specialized plant cells known as xylem vessels. Land plants must guard against drying out and so have evolved specialized structure known as stomata to allow gas to enter and leave the leaf. Carbon-dioxide cannot pass through the protective waxy layer covering the leaf (cuticle), but it can enter the leaf through stomata (the singular of stomata), flanked by two guard cells. Likewise, oxygen produced during photosynthesis can only pass out of the plant, while these gases are moving between the inside and outside of the leaf, a great deal of water is also lost. Cottonwood trees, for example, will lose 100 gallons (about 450dm³) of water per hour during hot desert days.

STAGES OF PHOTOSYNTHESIS

When chlorophyll a absorbs light energy, an electron gains energy and is 'excited'. The excited electron is transferred to another molecule (called a primary electron) and has a positive charge.

Photoactivation of chlorophyll a results in the splitting of water molecules and the transfer of energy to ATP and reduced nicotinamide adenine dinucleotide phosphate (NADP). The chemical reactions involved include:

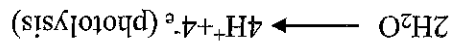
- Condensation reactions: responsible for water molecule splitting out, including phosphorylation (the addition of a phosphate group to an organic compound).
- Oxidation/reduction (redox) reactions involving electron transfer.

PROCESS OF PHOTOSYNTHESIS

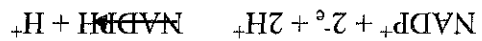
Photosynthesis is of two stage processes:

1. The light dependent reactions, a light dependent series of reactions which occur in the grana, and require the direct energy of light to make energy-carrier molecules that are used in the second process:

- Light energy is tapped by chlorophyll to make ATP (photophosphorylation).
- At the same time water is split into oxygen, hydrogen ions and free electrons



- The electrons then react with a carrier molecule changing it from its oxidized state (NADP⁺) to its reduced state (NADPH):



2. The light independent reactions, a light independent series of reactions which occur in the stroma of the chloroplasts, when the products of the light reaction, ATP and NADPH, are used to make carbohydrate from carbon dioxide (reduction), initially glyceraldehyde 3-phosphate (9 3-carbon atom molecule) is formed.

THE LIGHT-DEPENDENT REACTIONS

When light energy is absorbed by chlorophyll molecule its electrons gain energy and move to higher energy levels in the molecule (photo excitation). Sufficient energy ionizes the molecule, with the electron being 'freed' leaving a positively charged chlorophyll ion. This is called photoionization.

In whole chloroplasts each chlorophyll molecule is associated with an electron acceptor and an electron donor. These three molecules make up the core of photo system. Two electrons from photoionized chlorophyll molecule are transferred to the electron acceptor. The positively charged chlorophyll ion takes a pair of electrons from neighboring electron donor such as water.

THE LIGHT-INDEPENDENTS REACTION

In the light-independent process (the dark reaction) carbon dioxide from the atmosphere (or water for aquatic/marine organisms) is captured and modified by the addition of hydrogen to form carbohydrates. The incorporation of carbon dioxide into organic compounds is known as carbon fixation. The energy for this comes from the first phase of the photosynthesis process. Living system cannot directly utilize light energy, but can, through a complicated series of reactions, convert it into c-c bond energy that can be released by glycolysis and other metabolic processes.

Carbon dioxide combines with a five-carbon sugar, ribulose 1,5 biphosphate (RuBP). A six-carbon sugar forms but is instable. Each molecule breaks down to form two glycerat 3-phosphate (GP) molecules.

FACTORS AFFECTING THE RATE OF PHOTOSYNTHESIS

External factors

- **Light:** As light intensity increases, the rate of the light-dependent reaction, and therefore, photosynthesis generally, increases proportionately. As light intensity is increased, the rate of photosynthesis is eventually limited by some other factor. Low intensity of light causes the stomatal closure, which restricts the entry of carbon dioxide, resulting in the decrease of the photosynthesis. The wavelength (light quality) of light is also important. Photosynthesis occurs only in the visible part of the spectrum light, between wavelengths of 400nm and 700nm ie the photosynthetically active radiation (PAR). Light with a high proportion of energy concentrated in these wavelengths will produce a high rate of photosynthesis. Photosynthesis cannot take place either in the infrared or in the ultraviolet light. Light duration had significant influence on photosynthesis. Plant carries on a good rate of photosynthesis if it is given light for 10 to 12 hours per day.

- **Carbon Dioxide:** An increase in the carbon dioxide concentration increases the rate at which carbon is incorporated into carbohydrate in the light-independent reaction and so the rate of photosynthesis generally increases until limited by another factor.
- **Temperature:** Photosynthesis is dependent on temperature. It is a reaction catalyzed by enzymes. As the enzymes approach their optimum temperatures, the overall rate increases. Above the optimum temperature the rate begins to decrease until it stops

Internal factors

1. **Hydration of the protoplasm:** Hydration of the protoplasm is absolutely essential for almost all the vital processes of the plant, including photosynthesis. Decreased hydration of the protoplasm reduces the rate of photosynthesis.
2. **Chlorophyll content:** Chlorophyll in the green and yellow leaves varieties has little influence on the rate of photosynthesis. However, for unicellular green alga, there is a direct relationship between the chlorophyll content and rate of photosynthesis.
3. **Accumulation of end – products of photosynthesis:** If the rate of translocation of food, manufactured in the mesophyll cells, is not rapid enough with the rate of photosynthesis, the end products are bound to accumulate in the assimilatory cells resulting in a decrease in the rate of photosynthesis.